

Data Attribution and Sources

Primary Data

- V-Dem (Varieties of Democracy) Country-Year Full+Others, version 15 (2024 data release). Indicators used: v2cltrnslw, v2juhcind, v2juncind, v2clkill, v2cltort, v2xcl_acjst, v2x_freexp, v2x_clphy, v2clrspct.
- World Bank World Development Indicators (WDI) via API — GDP per capita (current US\$), population, prevalence of undernourishment, and poverty headcount at \$2.15 (2017 PPP). Values are the most recent published as of atlas production (labeled 2026 baseline).
- United Nations Member States list (193 sovereign states) for country scope and ISO3 codes.

Source Framework

The RTLDI equation, the nine RTLP indicators, and the nested causal interpretation are taken from: Sid J.A. Hubbard, Causality and Attraction: A Continuum of Steady States (Version 3, May 2026), DOI: 10.5281/zenodo.19468550. The operational crosswalk and binarization thresholds used here are documented in docs/indicator_crosswalk.md of this project.

This PDF was generated programmatically from the open RTLDI ATLAS toolkit. No proprietary or restricted data were used.

Nested Causal Modelling

The atlas is an application of nested causal modelling: the practice of identifying a small set of binary conditions whose presence or absence supports a target outcome (here, maximal GDP per capita). The nine levers are the conditions that emerged from modelling global data with that target. Their descriptions, costs, and multiplier effects are given in the front-matter section of the same name. The 25 % cap is a practical limiter on projections; the levers themselves are the transparent, low-cost, high-return part of the model.

Falsification of Malthusian Scarcity

Thomas Malthus argued that population grows exponentially while resources grow linearly, leading to famine and checks on population unless growth is deliberately limited.

The conditions Malthus projected are true when nations do not equally protect the lives of all people within their borders: more people then equate to less available resource per person. But when the right to life is equally protected, GDP rises for multiple reasons and rises exponentially because societal stresses are distributed evenly among a more productive and more capable society.

The levers make this possible at low annual cost. Freedom from torture, freedom from arbitrary detention, and an independent judiciary together make a nation investible — businesses and startups form when the risk of death or arbitrary imprisonment does not stop capital from flowing in. Whistleblower protections allow corruption and theft to be exposed at scales previously unknown. Most of the nine levers have minimal annual cost; they are force multipliers of economic velocity.

The important distinction is that this analysis does not negate the efficacy of Malthusian regimes. A nation can choose the Malthusian relationship to its population and receive the expected drain on resources (and plan for the expected trimming), or it can raise its RTLP score and enjoy the scaling benefits of an optimized enclosure. The cost of not caring for the lives of people equally is now visible: the choice is between expansion of corruption or stimulation of domestic industrial productivity.

Index of Terms

2026-fresh G0:

The most recent published GDP per capita values used as the dynamic baseline for the 2026 edition (even when RTLP components are from 2024).

Access to Justice:

Fair access to remedies for right-to-life violations

Arbitrary Detention:

Core protection against state overreach (RTLP indicator #4); failure contributes directly to attributable lost GDP in regions with weak rule of law.

Attributable lost GDP:

The share of a country or region's total lost output that is directly traceable to the absence of one specific RTLP indicator (sum of $\eta/9 \times G_0 \times \text{pop}$ over countries failing that indicator, with current $\eta \approx 0.30$ from population-weighted cross-section).

Binarization / threshold:

The documented conversion of continuous V-Dem and World Bank values into the binary 0/1 scores for the nine indicators (e.g., ≥ 2.0 on 0–4 scales, $\leq 5\%$ undernourishment + $\leq 10\%$ poverty for #9).

Causality and Attraction:

The source monograph (Hubbard 2026V3, Zenodo 19468550) that supplies the RTLDI equation, the nine RTLP indicators, the nested-enclosures paradigm, and the cartographic critique of distortion.

Choropleth:

Thematic map in which countries or regions are shaded by enclosure strength R (0–1) using the Viridis colormap; primary visual for the global, regional, and focus views.

Civilian Protection in Conflict Zones:

Mechanisms protecting civilians in conflict areas

Crosswalk:

The explicit mapping from the nine verbatim RTLP questions to observable V-Dem variables and World Bank statistics, with defined binarization thresholds.

ΔG (delta G):

Annual GDP per capita loss due to incomplete right-to-life protection. Core output of the RTLDI equation.

Diagnostic (tool/guide):

The use of RTLP scores, component breakdowns, and attributable lost-GDP figures as a practical instrument for citizens, NGOs, and policymakers to identify reform priorities and quantify upside.

Enclosure (nested causal / human):

The source document's central metaphor: the interdependent, layered structures of protection whose strength or weakness shapes higher-order economic and social steady states.

η (eta):

The sensitivity coefficient (current data-driven default 0.30 from population-weighted 2026 UN cross-section). In the bounded model $\Delta G = \min(\eta(1 - R) \times G_0, 0.25 \times G_0)$; the cross-sectional premium per indicator, then capped so the nine RTLP indicators are never credited with more than 25% of observed G_0 after other determinants are given due weight. (Source document used a more conservative 0.05 structural value.)

Existence of Legal Protections:

Provisions explicitly protecting the right to life

Fitbounds / regional zoom:

The per-nation choropleth view that automatically zooms to the target country plus the rest of its UN region, using the same Mollweide projection and Viridis scale as the global map.

Freedom from Torture and Inhumane Treatment:

Measures in place to prevent torture

Freedom of Expression and Whistleblower Protections:

Expression and whistleblower rights upheld

G_0 (G-zero):

Baseline GDP per capita (World Bank, current US\$). The 'current' economic size against which the deficit is measured.

Independent Judiciary:

Independent judiciary capable of upholding right-to-life laws

Law Enforcement Accountability:

Law enforcement agencies accountable for unlawful killings

Malthusian theory:

The proposition (Thomas Malthus) that population grows exponentially while resources grow linearly, inevitably producing scarcity and misery unless population is checked. The source and this atlas present geometric and empirical counter-evidence under conditions of equal right-to-life protection.

Member-nations table:

The compact table on each regional summary page that lists every country in the region together with its R, G0, population, and total lost GDP, plus a bottom row of regional totals.

Mollweide:

Equal-area pseudocylindrical map projection used for all choropleths (global, regional, and per-nation zooms) to ensure visual area corresponds to geographic reality.

Protection Against Arbitrary Detention:

Arbitrary detention prohibited with legal recourse

R (RTLP score):

Right-to-Life Protection score, 0–1. Average of nine binary indicators. 1 = full protection across all dimensions.

Right to Life:

The overarching protected interest; the nine indicators operationalize equal and effective protection against arbitrary interruption.

RTLDI:

Right-to-Life Deficit Index — the full framework and the per-country or global aggregate loss figure.

RTLP:

Right-to-Life Protection — the nine binary indicators that are averaged to produce R.

Socioeconomic Conditions:

Access to healthcare, food, and shelter adequate to sustain life

Steady state:

A balanced biological, economic, or social condition that the framework argues is supported or undermined by the strength of the nine life protections and the resulting causal enclosures.

Torture and Inhumane Treatment:

RTLP indicator #5; one of the largest contributors to global lost GDP, highlighting physical integrity failures.

Trend (3-year):

The per-nation GDP-driven RTLDI plots (R held fixed at 2024 values) that illustrate recent economic-drag dynamics alongside the regional and global maps.

UN region / regional summary:

One of the 22 geographic groupings used for aggregates, focused choropleths, member-nations tables (with R / G0 / population / lost + REGIONAL TOTAL row), and 9-indicator breakdowns.

V-Dem:

Varieties of Democracy project. Supplies the expert-coded governance and civil liberties indicators for eight of the nine RTLP components.

Viridis:

The perceptually uniform sequential colormap (dark low-R → yellow high-R) used consistently for every enclosure-strength choropleth in the atlas.

Credits and Acknowledgments

Conceptual Framework and Equations

Sid J.A. Hubbard — author of *Causality and Attraction: A Continuum of Steady States* (v3, 2026). All rights to the RTLDI equations, the nine RTLP indicators, and the nested causal paradigm remain with the author.

Data

- V-Dem Institute (<https://www.v-dem.net>) — V-Dem Country-Year dataset v15.
- World Bank Open Data (<https://data.worldbank.org>) — WDI via API.
- United Nations — list of 193 Member States.

Code and Atlas Production

RTLDI ATLAS project (2026). The open-source toolkit, crosswalk implementation, data pipeline, and this PDF generator are provided to make the framework usable by NGOs and researchers.

This work is released in the spirit of the original document: to map, with transparent data and reproducible math, the economic consequences of failing to protect the right to life equally.

For the full source code, 2026 data tables, and previous editions (2023, 2024), see the project repository. Users are encouraged to download the latest V-Dem and World Bank releases and re-run the pipeline for updated figures.